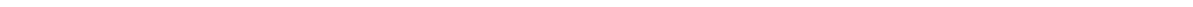




EX-EMPLOYEE VERIFICATION PORTAL

Project Documentation

A secure employee verification and appeals platform for organizations, verification agencies, and HR teams



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01. PROJECT AT A GLANCE

The **Ex-Employee Verification Portal** is a full-stack Next.js application designed to securely verify employment details of former employees.

The platform supports two primary user groups:

Verifiers

External companies or background verification agencies that submit employee verification requests.

Admin / HR Users

Internal HR or administrative users who review verification activity, handle appeals, inspect access logs and export verification data.

The application combines authentication, OTP-based login, employee verification, appeals handling, PDF report generation, email notifications, access auditing and supporting-document uploads into one centralized platform.

Core Capabilities

- Verifier Registration
 - Verifier Login
 - OTP-Based 2FA
 - Employee Verification
 - Verification Results
 - Match Score
 - Field-Level Comparison
 - Appeal Submission
 - Supporting Document Upload
 - Admin Dashboard
 - Appeal Management
 - Access Logs
 - Excel-Friendly Export
 - PDF Verification Reports
 - Email Notifications
 - Email Statistics
 - Session Timeout
 - Audit Logging
 - Role-Based Access
-

02. PROBLEM, USERS & PRODUCT VALUE

Problem

Employee verification is often handled through manual communication between companies, verification agencies and HR teams.

This can make it difficult to validate employee information, track verification requests, manage mismatches, handle appeals and maintain a reliable audit history.

The Ex-Employee Verification Portal centralizes these activities into one secure workflow.

Users

Verifiers

External organizations or background verification agencies that need to verify former employees.

They can:

Register

- Log in
- Complete OTP verification
- Submit employee details
- View verification results
- Raise appeals
- Upload supporting documents
- Generate verification reports

Admin / HR Users

Internal HR or administrative users who manage verification operations.

They can:

- View dashboard statistics
- Review appeals
- Inspect supporting documents
- Submit HR responses
- View access logs
- Export verification data
- Review email delivery statistics

Product Value

- Centralized verification workflow
 - OTP-based authentication
 - Employee data validation
 - Field-by-field comparison
 - Verification match scoring
 - Appeal management
 - Supporting document handling
 - PDF verification reports
 - Email notifications
 - Audit logging
 - Admin reporting
 - Secure role-based access
-

03. CUSTOMER PERSONA

Primary Persona: HR / Background Verification Manager

Goal: Review and respond to employee verification requests efficiently while maintaining a reliable record of every verification and appeal.

Needs

- Quickly review verification requests
- Validate employee information
- Identify mismatched fields
- Track verification results
- Review supporting documents
- Handle appeals
- Generate verification reports

Pain Points

- Manual verification processes
- Scattered employee information
- Difficulty tracking mismatches
- Unclear appeal status
- Manual document exchange

Ex-Employee Verification Portal Solution

The portal provides a centralized verification workflow where submitted employee information is validated against stored records, mismatches are identified, results are generated, appeals can be raised, and administrators can review the complete verification history.

04. ARCHITECTURE & TECHNOLOGY

The Ex-Employee Verification Portal uses a full-stack Next.js architecture with server-side API routes, MongoDB persistence and dedicated service layers.

Frontend

- Next.js 16
- App Router
- React 19
- Tailwind CSS 4
- DaisyUI
- Framer Motion

Backend

- Next.js API Routes
- Node.js
- JWT Authentication
- bcryptjs
- Joi Validation

Database

- MongoDB
- Mongoose

Email

- Nodemailer
- SMTP
- Gmail SMTP

File Storage

- Local filesystem in development
- Vercel Blob in production

Reporting

- jsPDF
- jsPDF-AutoTable
- html2canvas

Analytics

Recharts

Logging

Winston

Security

- Helmet
- CORS
- Rate Limiting
- JWT
- bcrypt
- Role-Based Access

Architecture

User → Next.js / React → API Routes → Services → Mongoose → MongoDB

The repository separates application pages, API routes, components, database utilities, models, services, hooks and validation logic.

05. CORE VERIFICATION WORKFLOW

The main workflow connects verifier authentication, employee validation, verification results, appeals and reporting.

Verifier Workflow

Registration → Login → OTP Verification → Verification Wizard → Employee Validation → Data Comparison → Verification Result → PDF / Appeal

Detailed Flow

- Verifier registers using a company email.
- Verifier logs in using email and password.
- A one-time password is sent for 2FA.
- After successful authentication, the verifier enters employee information.
- The system validates the employee ID and name.
- The backend compares the submitted information with stored employee records.
- A verification record is persisted.
- The result displays the match score and field-level comparison.
- If mismatches are found, the verifier can raise an appeal.
- Supporting documents can be uploaded with the appeal.
- A PDF verification report can also be generated.
- This workflow is documented as the core verifier experience.

Admin Workflow

Admin Login → Dashboard → Appeals / Logs → Review Appeal → HR Response → Export / Reporting

Admins can review pending and resolved appeals, inspect access logs, export verification data and review email delivery metrics.

06. FEATURE MAP

Landing Page

Introduces the employee verification platform and provides verifier registration access.

Verifier Registration

Collects:

- Company Name
- Company Email
- Password
- Background Verification Agency
- Personal email domains are rejected for verifier registration.

Verifier Login

Supports:

- Email
- Password
- OTP Verification
- Forgot Password
- Session Storage

Employee Verification

Multi-step verification wizard for submitting employee information.

Verification Comparison

Compares submitted information against employee records.

The comparison supports:

- Employee ID
- Employee Name
- Designation
- Exit Reason
- Entity Name
- Date Information
- The system calculates a match score and classifies the result as:
- Matched

- Partial Match
- Mismatch

Appeals

Verifiers can raise appeals when verification results contain mismatches.

They can:

- View mismatched fields
- Add comments
- Upload supporting documents
- Submit appeals

Admin Dashboard

Provides:

- Total Employees
- Total Verifiers
- Total Verifications
- Total Appeals
- Pending Appeals
- Recent Verification Activity
- Recent Appeal Activity
- Verification Status Breakdown

Access Logs

Tracks:

- Timestamp
- User Identifier
- Role
- Success / Failure
- IP Address
- Failure Reason
- User-Agent Information

Export

Admin users can export verification information through an Excel-friendly CSV download.

PDF Reports

Generate official verification reports from completed verification records.

Email Notifications

Supports:

- OTP Emails
- Welcome Emails
- Verification Reports
- Appeal Notifications
- Appeal Responses

File Uploads

Supporting documents can be uploaded during appeals. Development storage uses the local filesystem while production uses Vercel Blob.

07. BUILD: DESIGN MODE

Objective

Create a cohesive, high-fidelity UI design system for the **Ex-Employee Verification Portal** with a professional enterprise verification experience.

The actual Design Mode prompt defined five key screens and required them to follow the existing application's branding and color system.

Design Direction

- Forest Green primary branding
- Enterprise Blue secondary accent
- Pure White cards
- Soft Mint backgrounds
- Slate Gray body text
- Montserrat headings
- Lato body text
- Tailwind CSS
- DaisyUI
- Rounded cards
- Structured enterprise tables
- Clear step indicators
- Responsive layouts

Primary Brand

Forest Green: #007A3D

Used for:

- Headers
- Navigation
- Active steps
- Success badges
- Primary buttons

Secondary Accent

Enterprise Blue: #004F9E

Used for:

- Admin Login
- Secondary CTA
- Information tabs
- Highlight badges

Screens Designed

Screen 1

Verification Wizard Landing Page

[/verify](#)

Screen 2

Dual Authentication Portal

[/login](#)

[/admin/login](#)

Screen 3

HR Admin Management Dashboard

[/admin/dashboard](#)

Screen 4

Appeal Resolution & Document Review

[/admin/appeals](#)

Screen 5

Audit Log & PDF Certificate Generation

[/admin/logs](#)

These five screens and their detailed visual specifications are present in the actual Design Mode prompt sheet.

08. BUILD: PROTOTYPE MODE

Objective

Convert the approved Ex-Employee Verification Portal design into a functional full-stack application while maintaining the established enterprise visual system.

The implementation includes the public landing page, verifier authentication, verification wizard, admin dashboard and appeal-management experience.

Implementation

- Build / application implementation covered:
- Landing Page
- Verifier Registration
- Verifier Login
- OTP Verification
- Forgot Password
- Employee Verification Wizard
- Verification Result
- PDF Report
- Appeal Modal
- Admin Login
- Admin Dashboard
- Appeal Review
- Access Logs
- Export
- Email Statistics
- Settings / Shared Layout

Reusable Components

- Authentication Forms
- Verification Wizard
- Appeal Modal
- Admin Dashboard Cards
- Access Log List
- Excel Export Button
- Shared Header
- Shared Footer
- UI Components
- Forms
- Tables
- Status Badges

The project repository is organized around dedicated authentication, admin, verification, layout and UI components.

09. BUILD PROMPT HIGHLIGHTS

The Build prompts focused on creating a secure enterprise verification platform rather than a generic admin dashboard.

Main Prompt Focus

- Employee verification
- Verifier workflow
- Admin workflow
- OTP authentication
- Verification wizard
- Appeals
- Document review
- Audit logs
- PDF generation
- Responsive enterprise UI

Prompts Used

Prompt 1 : Verification Portal Design

Act as a Principal UI/UX Designer. Reverse-engineer and generate a cohesive, high-fidelity UI design kit with 5 key screens for the "Ex-Employee Verification Portal", matching the exact color scheme and branding of the existing web application. Use Forest Green (#007A3D) as the primary brand color, Enterprise Blue (#004F9E) as the secondary accent, white and soft mint backgrounds, Montserrat for headings and Lato for body text. Use Tailwind CSS and DaisyUI with rounded enterprise cards, clear step indicators and structured tables.

This is based directly on the actual Design Mode prompt sheet.

Prompt 2 : Verification Wizard

Create the Ex-Employee Verification Portal verification experience with a professional multi-step wizard. Include verifier information, employee search and verification result stages. Use clear step indicators, structured employee input fields, green focus states, validation feedback and a clear final result state. The interface should allow authenticated verifiers to submit employee details and understand the verification status and available next actions.

Prompt 3 : Admin Management

Create the HR Admin Management experience for the Ex-Employee Verification Portal. Include an admin dashboard with KPI cards for total verifications, pending appeals, approved cases and system logs. Add structured tables for verification activity, appeals and access logs, with clear status badges, filters and enterprise-style data presentation. Include actions for reviewing appeals, inspecting documents and exporting verification information.

The actual Design Mode specification includes these dashboard KPI cards, tables, status badges and admin workflows.

Prompt 4 : Appeals, Audit & Reporting

Extend the Ex-Employee Verification Portal with an appeal-resolution and reporting experience. Create a split-panel appeal review interface where HR users can inspect disputed verification requests, view uploaded supporting documents, review verifier comments and submit an approval or rejection response. Also provide an audit-log interface showing timestamps, user identity, action codes and IP information, together with an official PDF verification preview and export actions.

10. CORA: PLANNING & IMPLEMENTATION

The supplied project documentation shows a full implementation involving authentication, MongoDB, API routes, services, verification logic, appeals, PDF generation, email, file uploads and security.

The separate prompt-sheet file does not contain the original CORA prompts, so the prompts below are documentation-ready prompts based on the implementation documented in the project.

CORA was used for

- Backend API development
- MongoDB integration
- Authentication
- OTP system
- Verification logic
- Appeals
- PDF reports
- Email notifications
- File uploads
- Admin dashboard
- Access logging
- Validation
- Security

Workflow

Requirement → CORA Planning → Backend Implementation → API Integration → Testing → Verification

Prompts Used

Prompt 1 : Authentication & Session Architecture

Analyze the existing Ex-Employee Verification Portal authentication flow and implement a secure authentication architecture for both verifier and admin users. Use JWT-based authentication with bcrypt password hashing, protected API routes, role-specific session handling and proper token verification. Support verifier registration, login, OTP-based two-factor authentication, forgot-password flows and admin authentication while keeping the existing Next.js application structure intact. Ensure authenticated users cannot access protected verification or admin routes without valid authorization.

The implemented application uses JWT, bcryptjs, separate verifier/admin sessions and inactivity handling.

Prompt 2 : Employee Verification & Comparison

Implement the employee verification workflow for the Ex-Employee Verification Portal. Validate the submitted employee ID and name against the server-side employee records, verify company/entity consistency where applicable, compare the submitted information against stored employee data and persist the verification record. Implement field-level comparison for employee ID, designation, exit reason, entity name and relevant dates, including case-insensitive name comparison, date tolerance and match-score calculation. Return a clear verification classification of matched, partial_match or mismatch.

The documented comparison service implements these exact comparison rules and status classifications.

Prompt 3 : Appeals & Supporting Documents

Implement the appeal workflow for mismatched employee verification results. Require authenticated verifier access and ensure the verification record belongs to the requesting verifier. Collect all mismatched fields, allow the verifier to provide explanatory comments and optionally upload supporting documents with file type and size validation. Create a persistent appeal record and send an email notification to HR/admin. Implement the admin appeal-review flow so HR users can inspect metadata, comments and uploaded evidence and submit an approved or rejected response.

This matches the documented verifier and admin appeal workflow.

Prompt 4 : Admin Reporting, PDF & Audit

Implement the administrative reporting layer for the Ex-Employee Verification Portal. Provide dashboard statistics for employees, verifiers, verifications, appeals and verification status. Add access-log retrieval, Excel-friendly CSV export, email-delivery statistics and PDF verification-report generation. Ensure report generation verifies the verifier identity and record ownership before loading employee details and generating the report. Persist the generated report URL and return an existing report when one has already been generated.

These capabilities are documented across the admin dashboard, export, email statistics and PDF-report implementation.

11. CORA TASK ARCHIVE

Instead of listing every individual development task, the implementation can be grouped into the following major areas.

Authentication

- Registration
- Login
- OTP Verification
- Forgot Password
- JWT
- bcrypt Password Hashing
- Protected Routes
- Admin Authentication
- Verifier Authentication
- Session Validation
- Inactivity Timeout

Verification

- Employee Validation
- Employee ID Matching
- Name Matching
- Designation Matching
- Exit Reason Matching
- Entity Matching
- Date Comparison
- Match Score
- Verification Status
- Verification Record

Appeals

- Appeal Creation
- Mismatch Collection
- Verifier Comments
- Supporting Documents
- File Validation
- HR Review
- Approval / Rejection
- Appeal Notifications

Admin

- Dashboard Statistics
- Appeal Management
- Access Logs
- Data Export
- Email Statistics
- Admin Roles

Reports

- PDF Generation
- PDF Storage
- Report URL
- Report Email
- Existing Report Reuse

Data

- MongoDB
- Mongoose
- Employee Model
- Verifier Model
- Admin Model
- Verification Record
- Appeal
- Verification Attempt
- Access Log
- Email Log

Email

- OTP Emails
- Welcome Emails
- Verification Reports
- Appeal Notifications
- Appeal Responses

File Storage

- Local Uploads
 - Vercel Blob
 - File Type Validation
 - File Size Validation
 - Public URL
 - Storage Key
-

12. IMPLEMENTATION MILESTONES

01. Design

Created the Ex-Employee Verification Portal UI system using Build Design Mode.

02. Frontend

Implemented the landing page, authentication screens, verification wizard and admin interface.

03. Authentication

Implemented verifier and admin authentication using JWT, bcrypt and OTP-based verification.

04. Database

Connected MongoDB through Mongoose and established the required verification, user, appeal and audit models.

05. Verification

Implemented employee validation, field-level comparison, match scoring and verification status.

06. Appeals

Implemented verifier appeals, supporting-document uploads and HR/admin responses.

07. Reporting

Implemented PDF verification reports and Excel-friendly CSV exports.

08. Email

Implemented OTP, welcome, report and appeal-related email workflows using SMTP/Nodemailer.

09. Audit

Implemented access logs and verification activity tracking.

10. Security

Implemented JWT validation, password hashing, role-based route protection, OTP rate limiting and upload validation.

13. BACKEND, DATA & INTEGRATIONS

The application uses MongoDB with Mongoose as its primary persistence layer.

Core Data Models

- Employee
- Verifier
- Admin
- Verification Record
- Appeal
- Verification Attempt
- Access Log
- Email Log

Backend Flow

Frontend → Next.js API Route → Service Layer → Mongoose → MongoDB

Verification Flow

Verifier Input → Validation API → Employee Lookup → Comparison Service →
Verification Record → Result

Appeal Flow

Appeal → Validation → File Upload → Appeal Record → Email Notification → Admin Review

Report Flow

Verification Record → Ownership Check → PDF Generation → Report URL → Optional Email

Email Integration

The email subsystem uses Gmail SMTP through Nodemailer and logs email attempts into the database.

Supported email types include OTPs, welcome emails, verification reports, appeal notifications and appeal responses.

File Storage

Development → Local Filesystem

Production → Vercel Blob

Supporting documents are validated for file type and size before storage.

14. AUTHENTICATION & USER MANAGEMENT

The Ex-Employee Verification Portal uses JWT-based authentication with bcrypt password hashing and separate verifier/admin access flows.

Authentication Flow

Registration → Login → OTP → JWT Session → Protected Routes → Verification / Admin Dashboard

Verifier Authentication

- Company Name
- Company Email
- Password
- Verification Agency
- OTP Verification
- Forgot Password
- Session Storage

Admin Authentication

- Admin Login
- Admin-Specific API Path
- Admin Session
- Role-Based Access

Security

- JWT issuer and audience validation
- bcrypt password hashing
- Protected routes
- Role-based access
- OTP rate limiting
- Session inactivity timeout
- File validation
- Access logging

The documentation explicitly records JWT verification, bcrypt, role-based routes, OTP rate limiting and inactivity protection.

Session Management

Verifier session:

`verifier_session`

Admin session:

`admin_session`

Legacy fallback:

`auth_token`

The inactivity wrapper tracks activity, displays a warning, allows session extension and logs the user out after inactivity.

15. RESPONSIVE UI & POLISH

The Ex-Employee Verification Portal uses Tailwind CSS and DaisyUI layouts designed to scale across desktop, tablet and mobile.

UI Characteristics

- Enterprise visual language
- Forest Green branding
- Enterprise Blue accents
- Soft Mint backgrounds
- White cards
- Montserrat headings
- Lato body text
- Rounded components
- Structured tables
- Clear status badges
- Responsive forms
- Responsive modals
- Professional spacing
- Clear verification steps

The global design system uses Montserrat and Lato, a custom green-and-blue palette, DaisyUI theme overrides and responsive layouts.

Employee Verification

Verifier LoginAdmin Login

Secure & Instant Employee Verification

A fast, reliable, and automated platform for third-party verifiers to confirm employment history. Register your company to get started.

 Secure

 Fast

 Automated

Create a Verifier Account

Company Name *

Your Company Inc.

Your company's official name

Are you a background verification agency? *

☐ Yes

☐ No

Company Email Address *

verifier@yourcompany.com

Only company email domains are allowed (no Gmail, Yahoo, etc.)

Password *

Min. 6 characters

Minimum 6 characters long

Create Account

OR

Already have an account? [Login Now](#)

Made with CodeMate.ai

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Employee Verification

Verifier Login

Admin Login

Secure & Instant Employee Verification

A fast, reliable, and automated platform for third-party verifiers to confirm employment history.
Register your company to get started.

 Secure

 Fast

 Automated

Create a Verifier Account

Company Name *

Your Company Inc.

Your company's official name

Are you a background verification agency? *

☐ Yes

☐ No

Company Email Address *

verifier@yourcompany.com

Only company email domains are allowed (no Gmail, Yahoo, etc.)

Password *

Min. 6 characters

Minimum 6 characters long

Create Account

OR

Already have an account? [Login Now](#)

Made with CodeMate.ai

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Mobile

iPhone 12 Pro ▾ 390 x 844 76% ▾ No throttling ▾ 'Save-Data': default ▾

Emp Verification

Secure & Instant Employee Verification

Fast, automated employment verification for third-party verifiers. Register below to start.

 Secure

 Fast

 Automated

Create a Verifier Account

Company Name *

Your company's official name

BG verification agency? *

☐ Yes

☐ No

Company Email Address *

Company email required (no Gmail/Yahoo)

Password *

Minimum 6 characters long

Create Account

Made with [CodeMate.ai](#)

FINAL PRODUCT FLOW

Build Design Mode



Verification Portal UI



Build Code Mode



Authentication



OTP Verification



Employee Verification



Match Score & Result



Appeal + Supporting Documents



HR/Admin Review



PDF Report



Email Notification



Audit Logs & Export



Final Ex-Employee Verification Portal

